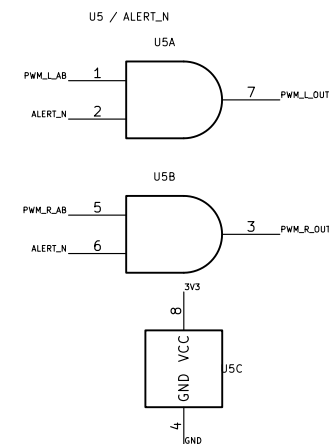
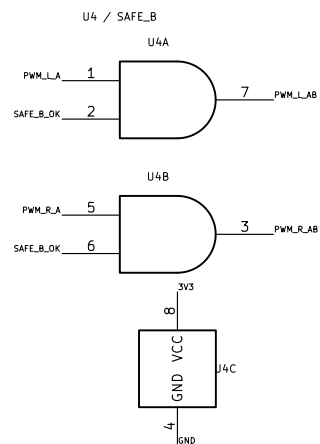
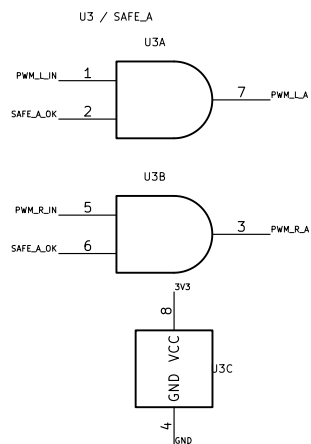
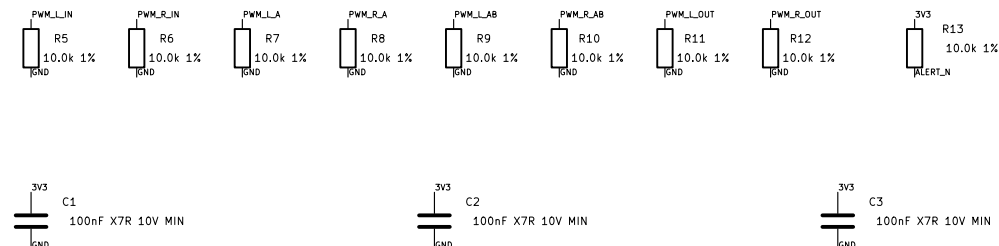
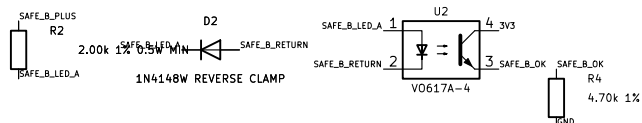
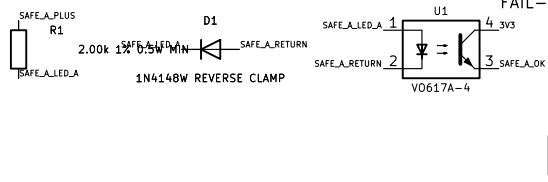


THREE-STAGE DUAL PWM PERMISSIVE CHAIN



FAIL-LOW BIAS, ALERT PULL-UP AND LOCAL DECOUPLING



CONNECTIVITY + ERC EVIDENCE ONLY. NO PCB, ROUTING, DRC, GERBER, ASSEMBLY, POWER-UP OR MOTION RELEASE.

MDD20A SIGN-MAGNITUDE MODE ONLY: PWM LOW IS BRAKE, NOT ENERGY ISOLATION. UPSTREAM CONTACTOR REMAINS REQUIRED.

SAFE_A, SAFE_B, ALERT_N, LOGIC-POWER LOSS AND STUCK-HIGH PWM REQUIRE <=20 ms BENCH WAVEFORMS BEFORE ACTUATOR POWER.

Sheet: /
File: stage19_dual_permissive_gate.kicad_sch

Size: A4	Date: 2026-07-15
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KiCad E.D.A. 10.0.4

Rev: S19-A

Id: 1/1